Lab report - Microscopy Imaging - How to build a virtual microscope

Group

November 9, 2026

1 Part 1 - Defining the Object

1.1 Q1

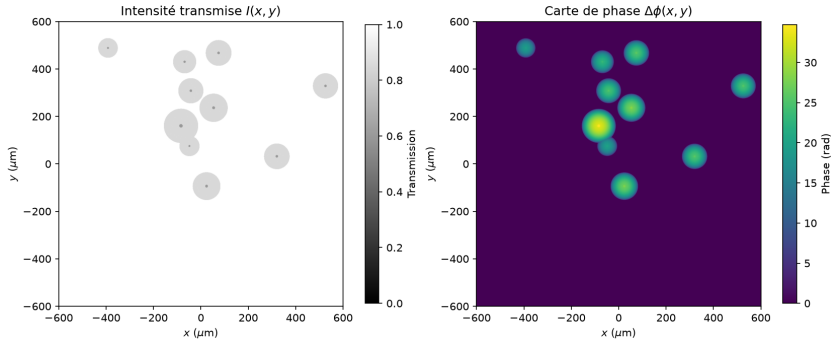


Figure 1: Synthetic object maps showing the transmitted intensity $I(x, y)$ and the accumulated phase delay $\Delta\phi(x, y)$ across a 1.2 mm $\times$ 1.2 mm field of view.

Q1.1 *What would the image of such an object look like on a traditional widefield microscope? What is the minimum numerical aperture that should allow you to resolve the nuclei?*

The image object is on an actual widefild microscope, a bunch of cells that have a certain thikness and absorb a certain amount of light. The image is then genuinely a phase object:

- $\Delta z = 0$ (focus): A camera or the eye detects only optical intensity $I = |E|^2 = 1$. The phase perturbations do not generate amplitude contrast when the object plane is conjugate to the lense (detector).
- $\Delta z \neq 0$ (out of contrast): As a transparent cell does not absorb light, but has a index of refraction that is a little bit higher than the fluid around, it acts as a small convergent lense, with a gradient index. So when the detector moves of Δz, the phase changes for the electromagnetic wave, and the focus is lost.

For the numerical apperture:

$$R_{\text{nuc}} = 50\,\mu\text{m} \times 0.10 = 5\,\mu\text{m}$$

$$\Leftrightarrow$$

$$D_{\text{nuc}} = 2 \times R_{\text{nuc}} = 10\,\mu\text{m}$$

$$\lambda = 550\mu m$$

We need to compute the diffraction limits. To resolve the object of a diameter of D_{nuc}, we must use two differents metrics:

- Abbe Resolution Limit:

$$d_{\text{min}} = \frac{\lambda}{2 \cdot NA} \leq D_{\text{nuc}} \implies NA \geq \frac{\lambda}{2D_{\text{nuc}}}$$

$$\Leftrightarrow$$

$$NA \geq \frac{0.55\,\mu\text{m}}{2 \times 10\,\mu\text{m}} = \mathbf{0.0275}$$

- Rayleigh Criterion:

$$r_{\text{Rayleigh}} = 0.61\frac{\lambda}{NA} \leq D_{\text{nuc}} \implies NA \geq 0.61\frac{\lambda}{D_{\text{nuc}}}$$

$$\Leftrightarrow$$

$$NA \geq 0.61 \times \frac{0.55\,\mu\text{m}}{10\,\mu\text{m}} \approx \mathbf{0.034}$$

So, the absolute theoretical lower bound to resolve the nucleus as an individual feature is $NA \approx 0.034$.

Q1.2 *What kind of technique could be used to retrieve information from a sample that is mostly transparent?*

There is two electromagnetic waves that goes through the sample:

- the direct one, that is not diffracted and has no phase lag;
- The one that goes across the cell and is diffracted by the nucleus ot the borders of the cell.

Thoses waves are in phase opposition. So a solution could be to set a dark-field: as dust particules lighten by the sun rays in a dark room, we can see very small dust particules that we would never without the darkfield. They are only visible because the diffraction of the light. This parallel is true in microscopy:

- We select only a cone of inclined light.
- We assert $NA_{illum} > NA_{obj}$: the angle of the light beam is superior to the objective aperture angle.
- When the light crosses a cell, the borders and organs diffract the light in evry directions.

Result: the background is dark, the cytoplasma is dark too (does not deflect light beams) but only the border is grey.

2 Part 2 - Resolution and the Pupil Function

Q3 *Using fft2, fftshift, ifft2, simulate imaging of the object for NA = 0.01, 0.1, 1.0. Compare the resulting images. Are the cells resolved? The nuclei?*

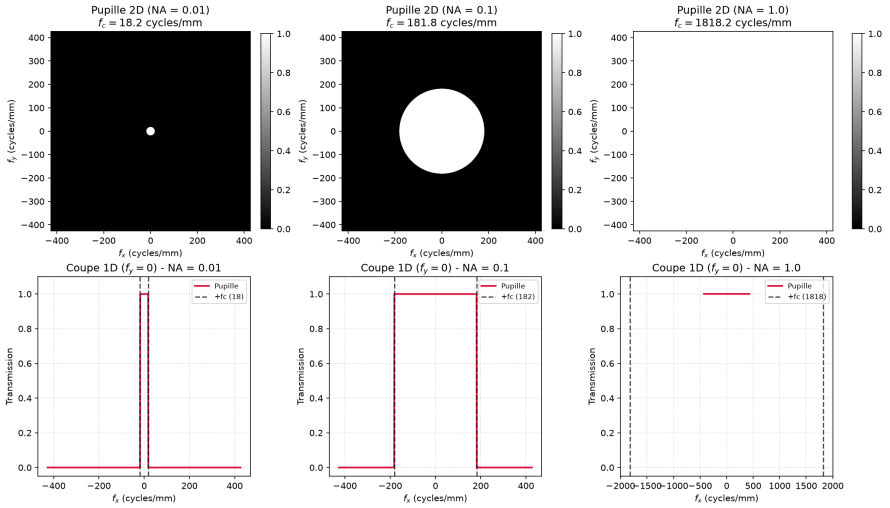


Figure 2: Pupil masks $P(f_x, f_y)$ and central 1D cross-sections for $NA \in \{0.01, 0.1, 1.0\}$, illustrating the spatial frequency cutoffs f_c acting as low-pass filters.

The coherent imaging process through a $4f$ optical telecentric system is simulated by convolving the complex field $E_{\text{in}} = \sqrt{I}e^{i\phi}$ with the objective Point Spread Function (PSF), implemented via the 2D Fourier transform:

$$E_{\text{out}}(x, y) = \mathcal{F}^{-1}\left\{\mathcal{F}\{E_{\text{in}}\}(f_x, f_y) \cdot P(f_x, f_y)\right\}, \quad I_{\text{det}} = |E_{\text{out}}|^2$$

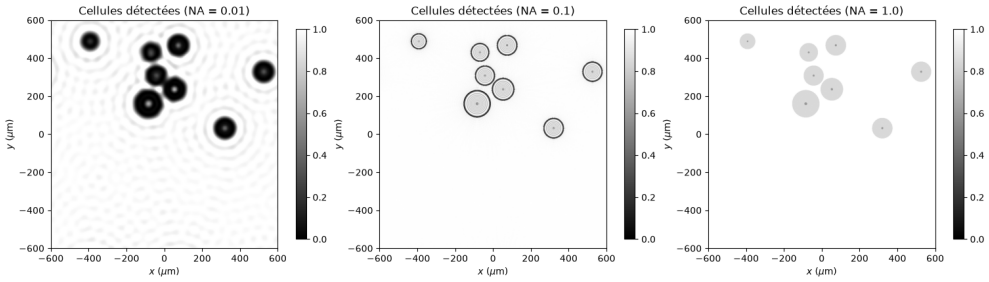


Figure 3: Simulated coherent imaging of the synthetic cell sample across three numerical apertures ($NA \in \{0.01, 0.1, 1.0\}$).

Comparing the three numerical apertures:

- $NA = 0.01$ **($r_{\text{Rayleigh}} = 33.6\,\mu\text{m}$):** Cutoff frequency ($f_c = 18.2$ mm^{-1}): restrictive. $r_{\text{Rayleigh}} \gg D_{\text{nuc}}$, so the nucleus vanishes into blurry dark stain. Cells overlap creates huges diffraction rings with no distinct boundaries.

- $NA = 0.10$ **($r_{\text{Rayleigh}} = 3.35\,\mu\text{m}$):** Since $r_{\text{Rayleigh}} < D_{\text{nuc}} = 10\,\mu$m, Rayleigh criteria are satisfied, and nuclei are distinctly resolved and cellular boundaries are clearly separable.

- $NA = 1.00$ **($r_{\text{Rayleigh}} = 0.34\,\mu\text{m}$):** Diffraction-limited resolution is achieved. Cutoff exceeds main spatial frequencies of the sample ($f_c = 1818$ mm^{-1}), reproducing cell contours and nuclear geometries with nearly perfect accuracy.

Q3.1 *Where is the pupil of the objective physically located?*

The back focal plane is behind the lense (objective). Though, as all the rays comes from the infinity, all parallal rays emmited by the sample ver a certain angle θ converges to a special point on this back focal plan (BFP). It is a physical apperture that truncate the angles accepted and foxes the maximal apperture angle: $\theta_{\max} = \arcsin(NA/n)$.

Q3.2 *Intuitively, how come the process of imaging through an optical system involves Fourier transforms?*

The light that goes through an object decomposes into a sum of plane waves. The angle of deflection then directly depends on the fine details. Indeed, the object's spatial frequencies are given by: $\sin\theta = \lambda f_x$. The lense focuses each wave Σ_θ into a single point x_f, y_f, in the BFP. This plane thus physically realizes a 2D spatial Fourier transform of the incident wave field. The second lens then collects these focal points and refocuses them onto the image plane, which performs the inverse operation (inverse Fourier transform-ifft/ifft2) of the recombined plane waves.

Q3.3 *A target can be used to quantify the resolution of a microscope. Use the different targets provided to try and assess the resolution of your system. Explain the method you are are using as well as your results.*

To calibrate optical resolution experimentally, a standard negative USAF 1951 resolution target is imaged through the pipeline for each numerical aperture:

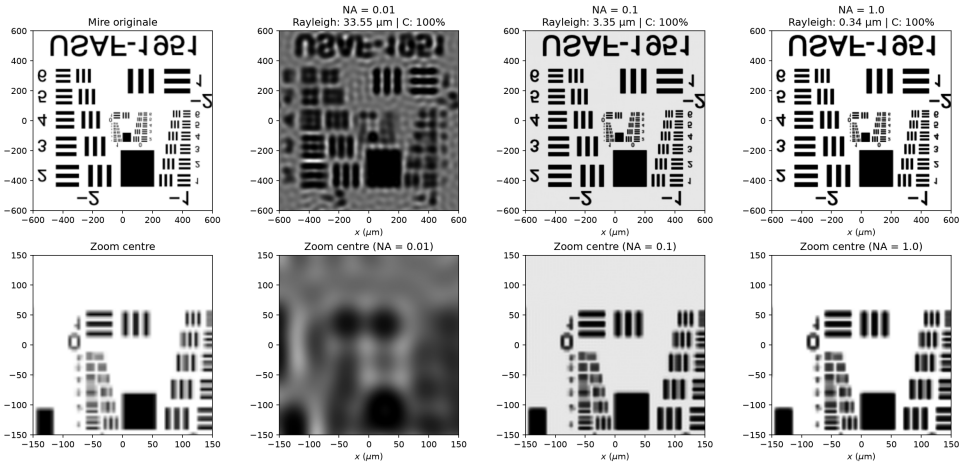


Figure 4: Simulation of the USAF 1951 resolution target across $NA \in \{0.01, 0.1, 1.0\}$, comparing global field views, central zoomed regions, and Michelson contrast modulation.

Methodology:

1. The target image is rescaled to physical coordinates ($L = 1.2$ mm, $N = 1024$), transformed to frequency space via `fft2`, filtered with the binary pupil mask, and back-transformed using `ifft2`.

2. A 1D intensity cross-section is extracted across the periodic bars in the central region of interest.

3. The contrast is evaluated across periodic elements:

$$C = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

   According to the Rayleigh criterion, two adjacent bars are considered resolved when the intensity dip between peaks is at least 19% to 26% ($C \geq 0.15$–0.20).

Observations

1. When $NA = 0.01$, the thick bars merges, the contrast decreases;

2. when $NA = 0.1$, the smallest element of the target we can see is the group 1, element 1. So, as : $Resolution = 2^{\left(\text{group number} + \frac{\text{element number} - 1}{6}\right)} = 2.00 lp/mm$ in line per milimeter. $d = \frac{1}{f} \times 1000 \quad (\mu\text{m})$, so $d = \mathbf{500}\,\mu\text{m}$.

3. when $NA = 1.0$, the smallest element of the target we can see is the group 1, element 2. So, as : $Resolution \approx 2.24 lp/mm$ in line per milimeter. So $d \approx \mathbf{445}\,\mu$m.

The natural choice for NA is then $NA = 0.1$ As the nucleus radius is about $10\mu m$, thus, it is never discernable.

3 Part 3 - Imaging Contrast Mechanisms

Q.4 In brightfield, the unscattered light dominates and the transparent cells are barely visible (only the absorbing nuclei and a faint edge). In darkfield, the annular pupil blocks the unscattered (zero-frequency) light: the background becomes black and only light scattered by the cells reaches the camera, so the cells appear bright, mostly at their edges where the phase gradient (light deflection) is strongest. Darkfield reveals the cell outlines but does not give a quantitative phase measurement.

Q5 *Phase contrast (weak object transfer function).*
For a weak phase object ($\varphi \ll 1$), the field can be linearized:

$$E = e^{i\varphi} \approx 1 + i\varphi.$$

The unscattered component (1) and the scattered component ($i\varphi$) are in quadrature, so $|1+i\varphi|^2 = 1+\varphi^2 \approx 1$: the cells are invisible. A Zernike phase plate multiplies the zero frequency by i (quarter-wave shift), giving

$$E' = i + i\varphi = i(1+\varphi) \quad \Rightarrow \quad I = (1+\varphi)^2 \approx 1 + 2\varphi,$$

so the phase is converted linearly into intensity. Since our cells reach $\varphi \approx 30$ rad, far from the weak-object regime, the phase was rescaled to $\varphi_{\max} = 0.3$ rad to stay within the validity of the linearization. In brightfield, this weak object is only visible through its absorption (panel 3), whereas in phase contrast (panel 4) the cells appear as bright disks whose brightness follows the phase map.

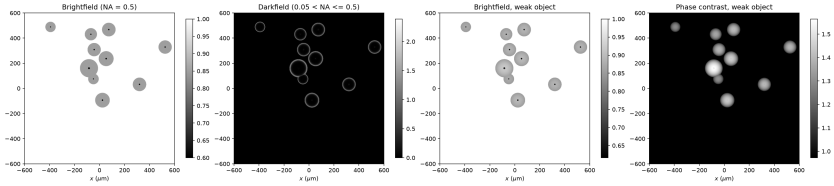


Figure 5: contrast images